

PROXY2WORLD: LEARNING TO GENERATE WORLDS FROM LIGHTWEIGHT PROXIES WITHOUT SEEING THEM

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ABSTRACT

Lightweight scene proxies let creators control scene layout and motion while leaving room for imagination in appearance, lighting, and visual effects. However, a suitable proxy is not uniquely defined, making paired proxy–video data difficult to construct automatically at scale. We present Proxy2World, a controllable world model that learns these complementary capabilities from ordinary posed RGBD videos, without training on authored proxy–video pairs. The model jointly learns depth-conditioned RGB generation and joint RGBD generation through cross-modal flow matching. Learning both tasks enables proxy–camera hybrid denoising at inference to follow the proxy structure while producing natural, detailed visuals. We further introduce ProxyBench to evaluate this capability across a diverse set of scenes, camera trajectories, and subject motions. Experiments on ProxyBench show that Proxy2World achieves a better balance between structural adherence and visual quality than camera-controlled and geometry-conditioned methods, supported by quantitative metrics, VLM assessments, human evaluations and diverse qualitative results. *Project page:* <https://dumdungura.github.io/proxy2world/>



Figure 1: **Build the structure. Generate the world.** Proxy2World transforms lightweight, interactive scene proxies into visually rich, structure-aligned worlds without paired proxy–video training data. Creators specify layouts and interactions through editable proxies (left), while the model generates detailed visuals (center) and variations in style, character, and lighting (right).

1 INTRODUCTION

Creating worlds that users can explore and direct is a central goal in gaming, filmmaking, and embodied AI. Recent models support action-driven interaction and camera-controlled exploration (Bruce et al., 2024; Che et al., 2025; He et al., 2025; Shen et al., 2026), while generative rendering translates explicit scene states into realistic imagery (Gu et al., 2025; Liang et al., 2025;

Gómez-Nogales et al., 2026; Chen et al., 2026b). For practical use, visual realism must be accompanied by control over scene layout, camera motion, and object movement. Yet specifying these properties should not require constructing every geometric and visual detail of the final world.

Existing approaches provide different levels of spatial control. Camera-controlled video models, including CameraCtrl, SCoPE, and GEN3C, guide viewpoint changes through pose conditioning, ray-based representations, or reconstructed scene caches (He et al., 2024; Yin et al., 2026; Ren et al., 2025). These methods enable controlled exploration, but a camera trajectory alone does not specify the desired scene layout or object motion. Geometry-aware video models introduce additional spatial structure. VACE and Cosmos-Transfer1 use depth or other spatial signals to guide video synthesis (Jiang et al., 2025; Alhaija et al., 2025), while DAR and UnividX use G-Buffer to connect editable scene states to generated observations (Chen et al., 2026b;a). World-consistent Video Diffusion, FantasyWorld, and Gen3R further couple visual generation with geometry, learning to generate appearance and spatial structure together (Zhang et al., 2025; Dai et al., 2026; Huang et al., 2026a). Video world models draw on learned priors to synthesize rich scenes and dynamics from sparse inputs, while generative rendering offers direct control through explicit scene representations. Bringing these capabilities together would allow users to prescribe scene structure and motion while leaving their visual realization to the generative model.

This raises a fundamental question: *how much of a world must be specified before generation can take over?* We study how lightweight scene proxies can guide world generation without specifying every detail of the final scene. A proxy may use low-poly meshes or simple primitives and may omit parts of the scene; the goal is to follow its intended layout and motion while generating plausible geometry and appearance beyond it.

Learning to generate from such proxies, however, presents several coupled challenges. First, a scene proxy reflects a designer’s abstraction of the world—which objects should be represented, how strongly they should be simplified, and which geometry can safely be omitted—so large-scale paired proxy-to-RGB data are difficult to obtain automatically.

Second, structural adherence must coexist with geometric refinement (Yan et al., 2026). A primitive specifies where an object is and how much space it occupies, but its simplified shape should be refined rather than reproduced exactly. Recent work C2R learns control using paired synthetic coarse-to-real examples (Gómez-Nogales et al., 2026); we instead aim to learn from ordinary RGBD videos and transferring that control to proxies at different levels of abstraction. Together, these challenges raise our central question: *can proxy-based control be learned without any paired proxy-to-RGB training data and adapt to different level of abstraction?*

We answer this question with Proxy2World, a controllable world model that learns proxy-based control from ordinary posed RGBD videos without paired proxy–video supervision. As illustrated in Figure 1, it turns editable scene proxies into visually rich worlds while allowing variations in style, character, and lighting. Our key insight is to jointly learn depth-conditioned RGB generation and joint RGBD generation through cross-modal flow matching. Learning both tasks enables proxy–camera hybrid denoising at inference to follow the proxy structure while producing natural, detailed visuals. We retain camera guidance throughout and align camera translations with the metric scale of depth to keep the two control signals consistent. We further introduce Proxy-Bench to evaluate proxy adherence, generative refinement, and camera control across diverse scenes, proxy abstractions, camera trajectories, and subject motions. Experiments show that Proxy2World achieves a better balance between structural adherence and visual quality than camera-controlled and geometry-conditioned methods, supported by operator-based metrics, VLM assessments, and human evaluations.

Our contributions are threefold:

1. **Proxy-based world generation without paired proxy supervision.** We introduce Proxy2World, a controllable world model that learns structural grounding and generative completion from ordinary posed RGBD videos, without paired proxy–RGB training data.
2. **Cross-Modal hybrid flow matching.** We combine proxy-grounded structure formation with camera-guided joint RGBD refinement, aligning geometry and camera motion in a shared metric scale to preserve the intended layout while refining geometry and appearance.

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3. **ProxyBench and comprehensive evaluation.** We introduce **ProxyBench**, an agent-driven benchmark spanning diverse scenes, camera trajectories, subject motions, and proxy abstractions. Evaluations using operator-based metrics, VLM assessments, and human preferences show that Proxy2World achieves a better balance between structural adherence and visual quality than camera-controlled and geometry-conditioned baselines.

2 RELATED WORK

Video foundations such as Stable Video Diffusion, CogVideoX, HunyuanVideo, and Wan provide generative priors for conditional synthesis (Blattmann et al., 2023; Yang et al., 2025; Kong et al., 2024; Wan et al., 2025). We review how subsequent methods expose camera, geometry, and scene-state controls, focusing on their relationship to generation from authored proxies.

2.1 CAMERA-CONTROLLED VIDEO GENERATION

Recent video diffusion models have substantially improved explicit camera control. MotionCtrl separates camera and object motion control (Wang et al., 2023b), while CameraCtrl and VD3D inject camera representations into pretrained video models (He et al., 2024; Bahmani et al., 2025). CamCo and CamI2V use epipolar constraints to structure cross-frame interactions (Xu et al., 2024; Zheng et al., 2024). CameraCtrl II extends camera-driven generation to dynamic scene exploration across wider viewpoints (He et al., 2025), and SCoPE incorporates camera sightlines into diffusion-transformer attention (Yin et al., 2026). These methods improve how a generator follows a requested viewpoint sequence. Geometric reprojection provides a complementary control interface. ViewCrafter uses point-based scene clues for novel-view generation (Yu et al., 2024), and GEN3C renders a reconstructed 3D cache along target camera trajectories (Ren et al., 2025). TrajectoryCrafter combines point-cloud renders with a source video to redirect its camera path (Yu et al., 2025). CamTrol obtains training-free camera control by using 3D layout rearrangement to guide noisy latents (Hou & Chen, 2024). CamCtrl3D combines pose, ray, reprojection, and 3D feature conditions (Popov et al., 2025), while RealCam-I2V aligns camera parameters with metric scene depth and applies scene-constrained noise shaping (Li et al., 2025). Proxy2World also couples geometry and camera scale, but accepts an externally authored scene proxy whose layout need not be reconstructed from the reference imagery.

2.2 GEOMETRY-AWARE VIDEO WORLD MODELS

Geometry can guide video synthesis as an input condition or be generated jointly with appearance. DiffusionRenderer learns rendering through G-buffers (Liang et al., 2025), while DaS and DAR synthesize video from mesh-derived controls (Gu et al., 2025; Chen et al., 2026b). VideoComposer, Control-A-Video, VACE, and Cosmos-Transfer1 support depth or other structural conditions (Wang et al., 2023a; Chen et al., 2023; Jiang et al., 2025; Alhaija et al., 2025), complemented by trajectory control in DragNUWA, Motion-I2V, and DragAnything (Yin et al., 2023; Shi et al., 2024; Wu et al., 2024). Geometry Forcing and GeoVideo improve geometric consistency (Wu et al., 2025; Bai et al., 2026). WVD, Aether, Voyager, FantasyWorld, Gen3R, VideoWeave, and DualCamCtrl further couple visual and geometric generation (Zhang et al., 2025; Zhu et al., 2025; Huang et al., 2025b; Dai et al., 2026; Huang et al., 2026a; Xiang et al., 2026; Zhang et al., 2026). We combine depth-conditioned grounding with joint RGBD refinement to follow abstract proxies while allowing their geometry to evolve.

2.3 PROXY-TO-RGB GENERATION.

C2R learns coarse-simulation control from paired synthetic examples (Gómez-Nogales et al., 2026). Concurrent work CWM and PWM separate programmable world states from visual synthesis, training their renderers on paired proxy-video or structured-control-video data (Chen et al., 2026c; Huang et al., 2026b). Marionette similarly renders predicted articulated states into pose controls for RGB generation (Meng et al., 2026). Proxy2World instead learns from ordinary posed RGBD videos, without paired proxy-RGB training data. Proxy-camera hybrid denoising combines structural grounding with joint geometry-appearance refinement, enabling transfer to low-poly, primitive, and incomplete proxies.

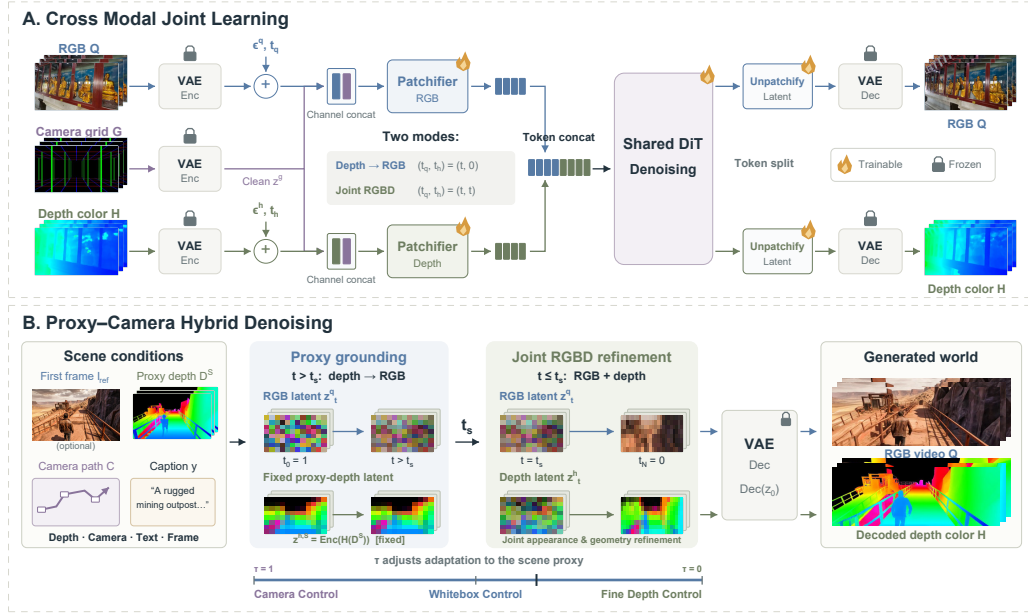


Figure 2: **Overview of the Method.** (A) **Cross Modal Joint learning.** We use a shared DiT to learn depth-conditioned RGB generation and joint RGBD generation, with camera-grid conditioning for both tasks. We train on ordinary posed RGBD videos without paired proxy-video supervision. (B) **Proxy-camera hybrid denoising.** We combine these learned capabilities during sampling: we first use proxy depth to establish scene structure, then jointly refine RGB and depth to enrich geometry and appearance. We retain camera-grid conditioning throughout. The refinement ratio τ controls the balance between proxy adherence and generative refinement.

3 METHOD

3.1 PROBLEM FORMULATION

Given an interactive scene proxy S , a control sequence $\mathcal{A}$, and a camera trajectory $\mathcal{C} = \{(\mathbf{K}_i, \mathbf{R}_i, \mathbf{t}_i)\}_{i=1}^F$, we render a proxy depth sequence:

$$\mathbf{D}^S = \{D_i^S\}_{i=1}^F = \mathcal{R}_{\text{depth}}(S, \mathcal{A}, \mathcal{C}), \quad (1)$$

where F is the number of frames, and $\mathbf{K}_i$, $\mathbf{R}_i$, and $\mathbf{t}_i$ denote camera intrinsics, rotation, and translation, respectively. The controls drive subject motion and scene interactions, which are conveyed to the generative model through rendered depth. Given a text description y and an optional reference image $\mathbf{I}_{\text{ref}}$ spatially aligned with the initial proxy view, our goal is to generate an RGB video:

$$\mathbf{Q} = \{\mathbf{Q}_i\}_{i=1}^F \sim p_{\theta}(\mathbf{Q} \mid \mathbf{D}^S, \mathcal{C}, \mathbf{I}_{\text{ref}}, y). \quad (2)$$

This task requires balancing structural adherence with generative freedom: strict conditioning can reproduce coarse proxy artifacts, while unconstrained generation can lose the intended layout and behavior. We seek to preserve proxy intent while refining geometry and appearance, learning from ordinary RGBD videos without paired proxy-to-RGB supervision.

3.2 SCENE PROXIES FOR EXPLICIT WORLD CONTROL

A scene proxy specifies control-relevant structure and behavior without prescribing the world’s final visual realization. Its geometry may be simplified or incomplete, preserving layout, occupancy, and subject motion while leaving visual details to the generative prior. We represent camera motion and scene geometry separately, supporting camera-only generation and additional structural control through proxy depth.

Camera grid. Following OmniDirector (Liu et al., 2026), we represent camera motion by rendering a fixed spatial grid $\mathcal{G}$ along the prescribed trajectory:

$$\mathbf{G} = \mathcal{R}_{\text{grid}}(\mathcal{G}, \mathcal{C}). \quad (3)$$

The grid provides spatial references whose projected motion expresses camera movement without specifying the target scene’s surfaces or appearance.

Metric depth. We express video depth and camera translations in consistent metric units. For training sequences whose depth and camera estimates share an arbitrary scale (Yan et al., 2025), we use DA3 (Lin et al., 2025) to estimate first-frame metric depth and align the sequence depth to this reference. The resulting sequence-level scale factor s is applied to both depth and camera translations:

$$D_i^{\text{metric}} = sD_i, \quad \mathbf{t}_i^{\text{metric}} = s\mathbf{t}_i. \quad (4)$$

This preserves their relative geometry while anchoring both to a common physical scale. Authored proxy depth and camera trajectories are likewise expressed in consistent metric units.

Following Vision Banana (Gabeur et al., 2026), we convert metric depth into a three-channel false-color representation:

$$\mathbf{H}_i = \mathcal{H}(D_i^{\text{metric}}) = h(f(D_i^{\text{metric}})), \quad (5)$$

where f nonlinearly maps metric distances to a bounded interval and h maps the result along a Hilbert-like path on the RGB cube. The nonlinear transform allocates greater precision to nearby geometry, while the invertible color mapping enables recovery of metric depth. A fixed mapping across frames and scenes preserves scale information and provides a common representation for observed depth and rendered proxy geometry.

3.3 CROSS-MODAL JOINT LEARNING

We jointly train depth-to-RGB generation and joint RGBD generation within a single flow-matching model. Both tasks are sampled throughout training and share the same parameters, allowing depth to serve as either a fixed geometric condition or a generated variable.

Shared architecture. As shown in Figure 2(A), a frozen video VAE encoder Enc maps RGB, metric-depth color, and camera-grid videos to latents $\mathbf{z}_0^q$, $\mathbf{z}_0^h$, and $\mathbf{z}^g$, respectively. The subscript 0 denotes clean data. Each modality latent is concatenated channel-wise with the clean camera latent and tokenized by a separate patch embedding. RGB and depth tokens are then concatenated along the sequence dimension and processed by a shared DiT, which predicts modality-specific velocities:

$$(\mathbf{v}_\theta^q, \mathbf{v}_\theta^h) = \mathbf{v}_\theta(\mathbf{z}_{t_q}^q, \mathbf{z}_{t_h}^h, t_q, t_h; \mathbf{z}^g, \mathbf{c}), \quad (6)$$

where t_q and t_h are modality-specific noise levels, and $\mathbf{c}$ contains text and optional reference-frame conditions. After sampling, the frozen decoder Dec recovers RGB and depth-color videos. We train the patch embeddings, shared DiT, and output projections.

Cross-modal flow matching. For each modality $m \in \{q, h\}$, we construct a noisy latent by interpolating between the clean latent $\mathbf{z}_0^m$ and independent standard Gaussian noise:

$$\mathbf{z}_{t_m}^m = (1 - t_m)\mathbf{z}_0^m + t_m\boldsymbol{\epsilon}^m, \quad \boldsymbol{\epsilon}^m \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (7)$$

where $t_m \in [0, 1]$, with 0 denoting clean data and 1 denoting pure noise. The target velocity is $\mathbf{u}^m = \boldsymbol{\epsilon}^m - \mathbf{z}_0^m$. For depth-conditioned RGB generation, we set $(t_q, t_h) = (t, 0)$: depth remains clean and only RGB is supervised. For joint RGBD generation, we set $(t_q, t_h) = (t, t)$ and supervise both modalities. Both tasks share the same model and are optimized through:

$$\mathcal{L} = \mathbb{E} \left[\sum_{m \in \{q, h\}} w_m \left\| \mathbf{v}_\theta^m \left(\mathbf{z}_{t_q}^q, \mathbf{z}_{t_h}^h, t_q, t_h; \mathbf{z}^g, \mathbf{c} \right) - \mathbf{u}^m \right\|_2^2 \right]. \quad (8)$$

The expectation covers task sampling, training data, noise levels, and Gaussian noise. We use $(w_q, w_h) = (1, 0)$ for depth-conditioned RGB generation and $(w_q, w_h) = (1, 1)$ for joint RGBD generation. This joint training allows the same model to use depth as a fixed structural condition or generate it together with RGB, providing the two capabilities combined during hybrid denoising.

3.4 PROXY-CAMERA HYBRID DENOISING

We compose the two learned generation modes within a single sampling trajectory. As illustrated in Figure 2(B), early steps use proxy depth to establish scene structure, while later steps jointly refine RGB and depth. Camera-grid guidance is retained throughout both stages. Let $\tau \in [0, 1]$ denote the fraction of sampling steps allocated to joint RGBD refinement. For a sampling schedule $1 = t_0 > \dots > t_N = 0$, we switch at $k_s = \lfloor (1 - \tau)N \rfloor$, with noise level $t_s = t_{k_s}$.

Proxy grounding. We encode the proxy depth-color sequence into $\mathbf{z}^{h,S} = \text{Enc}(\mathcal{H}(\mathbf{D}^S))$, where $\mathcal{H}$ is applied frame-wise, and initialize the RGB latent with Gaussian noise. During the early, high-noise steps, proxy depth remains fixed as a clean condition, and the model updates RGB using its depth-to-RGB mode:

$$\frac{d\mathbf{z}_t^q}{dt} = \mathbf{v}_\theta^q(\mathbf{z}_t^q, \mathbf{z}^{h,S}, t, 0; \mathbf{z}^g, \mathbf{c}), \quad t > t_s. \quad (9)$$

Sampling proceeds from $t = 1$ toward $t = 0$. This stage grounds generation in the proxy’s layout, occupied regions, and subject motion.

Joint RGBD refinement. At $t = t_s$, we retain the current RGB latent and initialize the depth state by adding noise to the proxy latent at the switching noise level:

$$\mathbf{z}_{t_s}^h = (1 - t_s)\mathbf{z}^{h,S} + t_s\epsilon^h, \quad \epsilon^h \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (10)$$

Depth then becomes a generated variable, and both modalities evolve under the joint RGBD mode:

$$\frac{d\mathbf{z}_t^m}{dt} = \mathbf{v}_\theta^m(\mathbf{z}_t^q, \mathbf{z}_t^h, t, t; \mathbf{z}^g, \mathbf{c}), \quad m \in \{q, h\}, \quad t \leq t_s. \quad (11)$$

The proxy is no longer imposed as a fixed depth condition, allowing geometry and appearance to be refined together while camera guidance continues to constrain the viewpoint sequence.

The refinement fraction τ controls the balance between adherence and refinement. A smaller τ retains proxy grounding for more sampling steps, whereas a larger τ allocates more steps to joint refinement. At $\tau = 0$, depth remains fixed throughout sampling; at $\tau = 1$, both modalities start from noise, yielding camera-conditioned generation without proxy grounding. Intermediate settings combine the two capabilities using the same checkpoint, without additional proxy-specific training.

4 EXPERIMENTS

Implementation details. We initialize Proxy2World from Wan2.2-I2V-A14B (Wan et al., 2025) and train on 120,624 RGBD clips from DL3DV (Ling et al., 2024), RealEstate10K (Zhou et al., 2018), and gameplay dataset (Zhou et al., 2025), each with 81 frames at 480×832 and 16 FPS, without paired proxy-RGB supervision. We freeze the video VAE and text encoder and optimize patch embeddings, DiT blocks, and output projections using AdamW (lr 10^{-5} , weight decay 0.01) on 48 GPUs. High-/low-noise experts are trained for 50,000/35,000 steps, with equal sampling weights for depth-to-RGB and joint RGBD tasks and uniformly sampled discrete noise levels within each expert’s shifted-flow schedule. Depth-to-RGB supervises RGB only; joint RGBD assigns unit loss weights to both modalities. The encoded RGB reference and temporal mask are concatenated to both streams, with reference dropout 0.5. At inference, available reference images provide DA3-estimated metric depth for proxy alignment, with the same scale correction applied to camera translations. We use 40-step Euler sampling, flow shift 3.0, and CFG 3.5. The joint-refinement fraction τ determines the switch $k_s = \lfloor (1 - \tau)N \rfloor$ for $1 = t_0 > \dots > t_N = 0$. We set $\tau^* = 0.925$: three geometry-conditioned steps followed by 37 joint RGBD steps, initializing refinement by noising proxy depth to t_{k_s} .

4.1 EXPERIMENTAL SETUP

ProxyBench. We introduce ProxyBench to evaluate world generation from coarse scene proxies. It comprises 25 scenes—15 medium-scale and 10 large-scale—with 300 five-second video sequences. For evaluation, we select a subset of 78 sequences emphasizing interaction and balanced

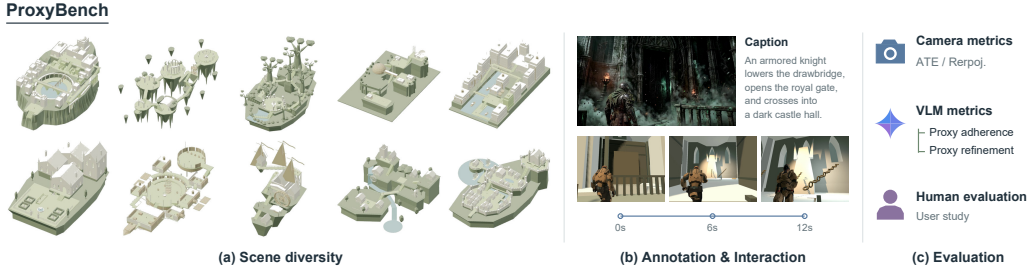


Figure 3: (a) Diverse lightweight scene proxies. (b) Text descriptions and scripted interactions specify the intended scene behavior and subject motion. (c) Evaluation combines camera and spatial consistency metrics, VLM assessments of proxy adherence and refinement, and human preferences.

Table 1: Evaluation on ProxyBench. Input icons indicate text, reference image, camera trajectory, and proxy control; gray icons denote unused inputs. Trajectory errors are computed from VIPE-estimated poses, and reprojection error excludes dynamic objects. Adherence and refinement are evaluated through Gemini pairwise win rates (%). Human ranks reflect overall preference. Bold and underline indicate the best and second-best results within each conditioning group. All our variants share one checkpoint trained without paired proxy- RGB data.

Method	Input	Camera		Spatial Cons. Reproj.↓	Proxy Evaluation					
		nATE _t ↓	nATE _r ↓		Operator-based		VLM-based		Human	
					Geo. Align↑	Temp. Align↑	Adh. WR↑	Ref. WR↑	Rank↓	
Camera-conditioned methods										
Lyra 2.0 (Shen et al., 2026)	T	0.0418	0.0413	<u>2.171</u>	<u>0.4965</u>	0.7068	32.21	39.48	10	
SCoPE / RAYPE (Yin et al., 2026)	T	0.1066	0.1229	4.338	0.4508	0.5780	29.51	47.06	9	
FantasyWorld (Dai et al., 2026)	T	0.2966	0.3472	2.449	0.4711	0.6158	<u>36.51</u>	<u>50.25</u>	7	
Ours (TI2V, $\tau = 1$)	T	<u>0.0681</u>	<u>0.0524</u>	1.956	0.5340	<u>0.7033</u>	53.55	71.44	5	
Depth-conditioned methods										
VACE-Depth (Jiang et al., 2025)	T	0.0764	0.0584	6.264	0.7576	0.7335	60.14	45.32	<u>4</u>	
Cosmos-Depth (Alhaija et al., 2025)	T	<u>0.0446</u>	<u>0.0362</u>	<u>3.259</u>	<u>0.7713</u>	<u>0.7504</u>	72.26	<u>40.28</u>	3	
Ours (T2V, $\tau = 0$)	T	0.0302	0.0270	2.280	0.7749	0.7645	<u>68.79</u>	31.71	6	
Proxy-conditioned methods										
Coarse2Real (Gómez-Nogales et al., 2026)	T	0.1637	0.3152	5.312	0.5520	0.6616	40.03	45.21	8	
Ours (T2V, $\tau = \tau^*$)	T	<u>0.0863</u>	<u>0.0719</u>	2.067	0.7084	<u>0.7126</u>	<u>42.81</u>	<u>51.30</u>	<u>2</u>	
Ours (TI2V, $\tau = \tau^*$)	T	0.0659	0.0558	<u>2.524</u>	<u>0.6795</u>	0.7173	64.18	77.96	1	

coverage of motion types. Each case provides a proxy scene, a target camera trajectory, synchronized proxy renderings, a text prompt, and a reference first frame. Successful generation should preserve the intended layout and behavior while enriching coarse geometry and appearance. Figure 3 provides an overview of ProxyBench, including scene diversity, interaction annotations, and the evaluation protocol.

Baselines. We select baselines covering geometry-conditioned rendering, camera-controlled world generation, and coarse-to-real synthesis. Among geometry-conditioned approaches, we focus on VACE (Jiang et al., 2025) and Cosmos-Transfer (Alhaija et al., 2025) as strong depth-conditioned video generators, adapting them to proxy control using depth rendered from our coarse scenes. For camera-controlled generation, Lyra 2.0 (Shen et al., 2026) represents projection-based approaches, but its static-scene formulation limits dynamic interactions. We therefore also include SCoPE/RayPE (Yin et al., 2026) for ray-based camera control and FantasyWorld (Dai et al., 2026) for joint RGBD world generation. Coarse2Real (Gómez-Nogales et al., 2026) serves as the most direct baseline, explicitly translating coarse simulations into realistic videos. Our camera-only ($\tau = 1$), depth-only ($\tau = 0$), and mixed-control ($\tau = \tau^*$) variants share one checkpoint. Table 1 specifies the inputs used by each configuration.

Evaluation Protocol. We evaluate camera control using normalized translation and rotation errors (nATE_t and nATE_r) computed from VIPE-estimated poses (Huang et al., 2025a). Spatial consis-

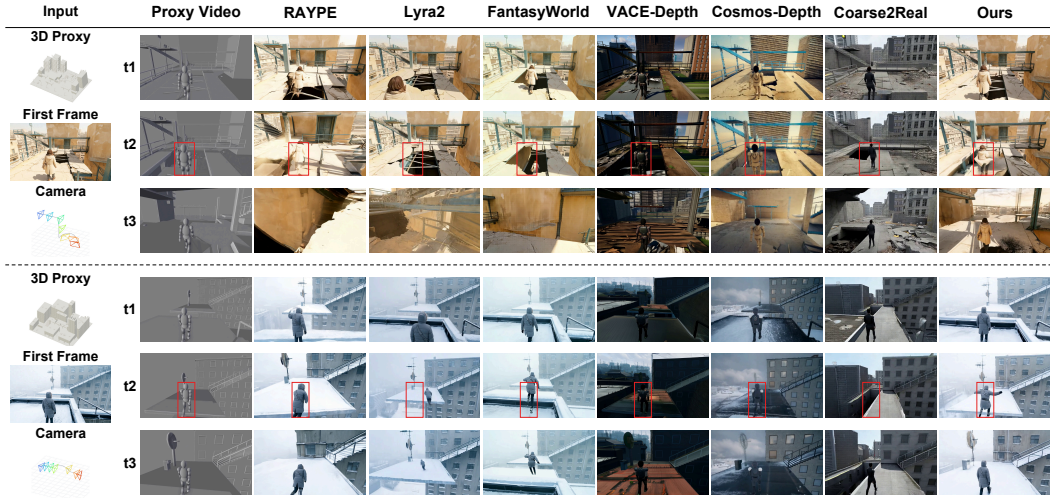


Figure 4: **Qualitative comparison on ProxyBench.** Red boxes highlight the intended subject locations specified by the proxy, highlighting differences in subject placement and shape. Camera-conditioned methods can deviate from the prescribed subject placement, while depth-conditioned methods can retain coarse body shapes and produced distorted characters. Ours preserves the intended placement while generating more natural character shapes and detailed scene appearance.

tency is measured by reprojection error after excluding dynamic objects. Geometric and temporal alignment measure agreement with the proxy. We further use Gemini-v3.7-flash(Team et al., 2023) for pairwise evaluation of *Proxy Adherence* and *Proxy Refinement*. Adherence assesses preservation of the intended layout and subject behavior, while refinement assesses plausible enrichment of geometry, appearance, and motion. We report average win rates across opponents, counting ties as half a win, alongside overall human preference ranks. We evaluate all methods on a common set of 78 sequences using operator-based metrics and VLM assessments, complemented by a human preference study with 10 participants on 10 sequences. Detailed protocols are provided in the appendix.

4.2 EVALUATION ON PROXYBENCH

Quantitative comparison. Table 1 shows that Proxy2World achieves a strong balance between proxy adherence and generative refinement. Compared with camera-controlled methods, our mixed TI2V model achieves higher geometric alignment and VLM adherence while also attaining the highest refinement win rate. The comparison with our own camera-only TI2V variant isolates the benefit of proxy guidance: geometric alignment improves by 27.2%, while adherence and refinement win rates increase by 10.63 and 6.52 percentage points, respectively. Thus, explicit proxy control improves structural adherence without sacrificing the model’s generative flexibility. Compared with depth-conditioned methods, mixed generation allows greater refinement of the supplied coarse geometry. Cosmos-Depth achieves stronger VLM adherence (72.26% vs. 64.18%), but our mixed TI2V model achieves a substantially higher refinement win rate (77.96% vs. 40.28%). Together with its first-place ranking in human preference, these results support the benefit of balancing structural constraints with the freedom to refine coarse inputs. Compared with the proxy-conditioned baseline Coarse2Real as the same proxy method, our mixed T2V model reduces translation and rotation errors by 47.3% and 77.2%, respectively, and improves geometric alignment by 28.3%. These gains are obtained without authored proxy–video training pairs, demonstrating that ordinary RGBD supervision can support effective structural control on authored proxies.

Qualitative comparison. Figure 4 highlights two common failure modes. Depth-conditioned methods follow the supplied structure but can reproduce the coarse proxy’s simplified body shapes, resulting in distorted characters and unnatural proportions. Camera-conditioned methods generate more natural-looking content, but can deviate from the proxy’s scene layout, subject placement, and motion. Proxy2World combines structural adherence with geometric refinement: it preserves the



Figure 5: Effect of the refinement ratio and model components. (a) Varying τ with all other inputs and the random seed fixed illustrates the trade-off between proxy adherence and refinement. (b) Ablation of metric scale alignment and joint RGBD learning, evaluated through camera accuracy and human preference.

prescribed spatial relationships and subject motion while producing more natural character shapes and detailed scene appearance. The highlighted regions illustrate this difference, showing how our method refines coarse subjects while retaining their intended placement within the scene.

4.3 ABLATION STUDIES

Hybrid denoising. Figure 5(a) examines how the two generation modes contribute to proxy-based control. Proxy-only generation preserves the prescribed layout and subject placement, but tends to carry simplified proxy shapes into the output, particularly for characters. Camera-only generation produces more natural shapes and richer appearance, but can alter subject placement and scene structure. Mixed generation preserves the major spatial relationships while refining coarse characters and adding scene details. These results support our use of early proxy grounding to establish structure and subsequent joint RGBD refinement to improve its visual realization, combining the strengths of both learned modes without additional proxy-specific training.

Metric scale alignment. We align depth and camera translations to a common metric scale across training scenes. Normalizing both consistently within each scene preserves their relative geometry, but does not establish a shared physical scale across scenes. Metric alignment therefore provides a consistent scale convention for learning from depth and camera-grid conditioning. Figure 5(b) shows that removing this alignment during training yields the largest camera error among the tested variants and substantially lowers human preference. These results highlight the importance of a shared metric scale across training scenes for accurate camera control and preferred visual results.

Joint RGBD learning. To test whether refinement requires joint geometry–appearance modeling, we replace joint RGBD generation with camera-conditioned RGB-only generation while retaining the initial depth-conditioned grounding stage. This variant still releases the fixed proxy-depth constraint during later sampling steps, but no longer generates depth alongside RGB. As shown in Figure 5(b), it yields higher camera error and lower human preference than the full model. The improvement therefore cannot be attributed solely to removing the depth constraint: explicitly generating geometry together with appearance contributes to both camera accuracy and visual quality. This supports joint RGBD learning as a key component of our refinement stage.

5 CONCLUSION

We presented Proxy2World, a unified RGBD world model that generates camera-controllable videos from lightweight proxies without paired proxy–video training data. Proxy–camera hybrid denoising combines structural grounding with joint RGBD refinement, balancing proxy adherence and visual quality as demonstrated on ProxyBench. Our approach lets creators specify coarse structure while leaving visual details to generation. Long-horizon interactive generation remains future work.

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